

SHOVON MANDAL

Lecturer, Department of Computer Science and Engineering | Northern University of Business and Technology Khulna, Bangladesh
Email: shovonmandal@gmail.com | [ResearchGate](#) | [Google Scholar](#) | [GitHub](#) | [LinkedIn](#) | [ORCID](#) | [Website](#) | Location: Bangladesh

RESEARCH PROFILE

Computer Science and Engineering lecturer and graduate researcher working on blockchain-based secure digital payment systems, NFC-enabled cashless transactions, and secure distributed architectures. Published peer-reviewed conference papers in blockchain-enabled payment systems, IOTA-based decentralized communication, machine learning, NLP, and AI-enabled applications. Seeking fully funded graduate research opportunities in the United States in blockchain security, IoT/NFC transaction security, applied cryptography, and secure financial technologies.

RESEARCH INTERESTS

Blockchain-Based Secure Digital Payments; NFC and IoT Transaction Security; Secure Distributed Systems; Smart Contracts; Applied Cryptography; Embedded Systems Security.

EDUCATION

M.Sc. in Computer Science and Engineering | *Khulna University of Engineering and Technology, Bangladesh* **Jul 2022 – Present**

Status: Defense Pending | Coursework CGPA: 3.29/4.00

Thesis: Blockchain Technology Based Smart E-Payment Strategy for Secure Cashless Payment

Supervisor: [Dr. Pintu Chandra Shill](#) | Research Area: Blockchain-Based Secure Digital Payment and NFC-Enabled Transaction

Relevant Coursework: Principles of Cryptography; Advanced Network Security; Cloud Computing; Neural Networks.

B.Sc. in Computer Science and Engineering | *BRAC University, Bangladesh*

Spring 2018 – Spring 2022

CGPA: 3.80/4.00 | Academic Standing: Highest Distinction

Thesis: Blockchain-based Micropayment System for Secured Cashless Small Payments

Supervisor: [Md. Golam Rabiul Alam, PhD](#) | Research Area: Blockchain-Based Micropayment and Secure Cashless Transactions

PEER-REVIEWED PUBLICATIONS

Published Conference Papers

- Mandal, S., & Shill, P. C.** (2024). "Ensuring Security and Efficiency in Digital Payments using Blockchain with NFC." *2024 International Conference on Knowledge Engineering and Communication Systems (ICKECS)* DOI: [10.1109/ICKECS61492.2024.10616703](https://doi.org/10.1109/ICKECS61492.2024.10616703).
- Ghosh, A., **Mandal, S.**, Islam, M. J., Islam, K. M., Dhar, S., & Baidya, R. (2026). "Comparative Evaluation of Multiple CNN Architectures for Dermoscopic Skin Lesion Classification using ISIC Dataset." *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)* DOI: [10.1109/QPAIN69676.2026.11545903](https://doi.org/10.1109/QPAIN69676.2026.11545903).
- Keya, M., **Mandal, S.**, et al. (2024). "Q-learning based Automated Message Multicast in Gossip Protocol for Node Confirmation in IOTA Tangle." *2024 27th International Conference on Computer and Information Technology (ICCIT)* DOI: [10.1109/ICCIT64611.2024.11021995](https://doi.org/10.1109/ICCIT64611.2024.11021995).
- Ghosh, A., Pinki, F. T., Baidya, R., Alam, F., & **Mandal, S.** (2026). "A Machine Learning and DEM-Based Approach for Predicting Sea Level Rise and Assessing Coastal Risk in Bangladesh Using AI Insights." *2026 IEEE 15th International Conference on Communication Systems and Network Technologies (CSNT)* DOI: [10.1109/CSNT69054.2026.11502358](https://doi.org/10.1109/CSNT69054.2026.11502358).

Accepted / In Press Conference Papers

- Mandal, S.**, et al. (2025). "Revolutionizing Academic Learning and Research with the Integration of Leap Motion and ARCore in Android." *16th International IEEE Conference on Computing, Communication and Networking Technologies (ICCCNT), IIT Indore, India* [In press].
- Ghosh, A., **Mandal, S.**, et al. (2025). "Advancing Sentiment Analysis Through Natural Language Processing and Deep Learning." *16th International IEEE Conference on Computing, Communication and Networking Technologies (ICCCNT), IIT Indore, India* [In press].

SELECTED RESEARCH AND DEVELOPMENT PROJECTS

Blockchain and NFC-Based Secure Cashless Payment Prototype | *Thesis-Supporting Research Implementation* | GitHub available

- Developed a secure cashless payment prototype integrating blockchain-based verification with NFC-enabled payment interaction.
- Implemented supporting Arduino and Python modules for NFC-based transaction processing and system-level experimentation.
- Repository:** github.com/shovon-mandal/NFC | Related Output: ICKECS 2024 paper on Blockchain with NFC.

IOTA-Based Payment Website and Decentralized Transaction Demonstration | *Web-Based Research Prototype* | GitHub available

- Developed a web-based prototype to demonstrate IOTA-supported payment interaction and decentralized transaction flow.
- Implemented front-end and backend components for payment demonstration, user interaction, and experimental validation.
- **Repository:** github.com/shovon-mandal/IOTA-Website | Related Output: ICCIT 2024 paper on IOTA Tangle.

ACADEMIC APPOINTMENTS AND TEACHING EXPERIENCE

Lecturer, Department of CSE | *Northern University of Business and Technology Khulna, Bangladesh* **Mar 2024 - Present**

- Teach undergraduate theory and laboratory courses in computer architecture, digital logic, object-oriented programming, technical writing, and optimization.
- Prepare lecture materials, laboratory tasks, assessments, review resources, and academic evaluation materials; support student mentoring and research engagement.

Courses Taught: Computer Architecture; Digital Logic Design; Object-Oriented Programming with Java; Technical Writing; Linear Programming and Combinatorial Optimization; Digital Logic Design Lab; Linear Programming and Combinatorial Optimization Lab

Lecturer (Adjunct), Department of CSE | *Imperial College of Engineering, Khulna, Bangladesh* **Jul 2022 - Jun 2024**

- Delivered undergraduate lectures, laboratory sessions, assessments, and academic support for core CSE courses.

Courses Taught: Database Management Systems; Computer Peripherals and Interfacing; Computer Architecture and Organization; Computer Graphics; Web Engineering; Microprocessor and Microcontrollers

Teaching Assistant, Department of CSE | *Khulna University of Engineering and Technology, Bangladesh* **Jul 2023 - Dec 2023**

- Assisted faculty members in laboratory instruction, assignment evaluation, academic record preparation, and student support.

Courses Assisted: Embedded Systems; Electronic Devices and Circuits

Student Tutor, Department of CSE | *BRAC University, Bangladesh* **Feb 2022 - Apr 2022**

- Provided individual and group academic support to undergraduate students through tutoring sessions and guided academic assistance.

ADDITIONAL TECHNICAL PROJECTS

Student Management Android Application | *Course Project* | GitHub available

- Developed an Android-based student management application using Android Studio. Repository: github.com/shovon-mandal/Student-Management
- Completed hardware-oriented prototypes including a traffic light system and a fire alarm system integrated with mobile SIM communication.

TECHNICAL SKILLS

Programming: Python, Java, C, MATLAB, MIPS Assembly, JavaScript

Web/Database: MySQL, PHP, Laravel, HTML, CSS, JavaScript

Research/Data Tools: LaTeX, SPSS, Microsoft Excel, Google Sheets, Git, API integration, workflow automation

Systems/Hardware: Arduino-based prototyping, NFC-based payment modules, embedded systems, hardware interfacing, digital electronics, Android Studio

Blockchain/Security: Blockchain-based payment prototyping, IOTA payment demonstration, applied cryptography concepts, secure distributed systems

Languages: Bangla (Native), English

HONORS, SCHOLARSHIPS, AND ACADEMIC SERVICE

- Highest Distinction, BRAC University.
- Vice Chancellor's List, BRAC University — 5 consecutive semesters.
- Dean's List, BRAC University — 2 consecutive semesters.
- Merit-based Scholarship, BRAC University.
- Executive Member, BRAC University Computer Club, 2018–2020.
- Executive Member, Robotics Club of BRAC University, 2018–2020.
- Executive Member, BRACU Mongol Tori, Next Generation Mars Rover Team 4.0, 2019–2020.
- References available upon request.

References and supporting documents are available upon request.